



BantayAralan: Real-Time Student Behavior and Classroom Orderliness System

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Classroom management is an important part of education, as how teachers manage their classrooms can influence students' academic success as well as their social development. Everyday, teachers face unique challenges when trying to manage or observe multiple students simultaneously, especially when managing large classes or students with varying attention spans and behavioral tendencies. Disruptive behaviors, if not addressed promptly, can affect not only the student engaged in the behavior but also their classmates, reducing overall classroom efficiency and learning experience of students.

Effective classroom management directly influences the students' academic performance. Research indicated that good management of classroom environments, a calendar for activities/releasing students from class, and a consistent approach to disciplinary practices were linked with higher performance (Çali et al., 2024). In comparison, students who behaved poorly exhibited decreases in academic achievement, including lower levels of participation in class and shorter time periods of attention while in class (Hamidi et al., 2024). The findings highlight how classroom management is directly connected with academic performance, influencing not only individual student outcomes but also overall classroom dynamics, teacher effectiveness, and the quality of instruction.

While classroom management is important, many teachers struggle balancing teaching responsibilities with managing disruptive student behavior. A study among 130 Philippine teachers found that handling student misbehavior, such as fighting and attention-seeking, is a frequent and ongoing issue, with only selected management strategies showing significant effectiveness (Dilabayan & Sambo, 2024). This indicates that even experienced teachers must constantly divide their attention between instruction and behavior control. Additionally, repetitive administrative tasks and paperwork further distract teachers from classroom duties,

reducing instructional time and increasing stress (Magsambol, 2025). These factors collectively highlight the need for systems or interventions that can support teachers in both managing student behavior and delivering quality instruction.

Aside from monitoring student action, classroom management also means ensuring a clean classroom in which teachers in the Philippines also encourage students to practice cleanliness and orderliness as part of student discipline. Clean, properly organized classrooms are essential for effective learning, according to the Department of Education (2023). Teachers are expected to assist students with keeping classrooms clean. Many of these items remain in classrooms at the completion of class and contribute to cluttering the environment. Reyes et al. (2024) discussed in their recent research study that 49% of the respondents believed their school's cleanliness did not have an impact on the classroom environment; therefore, the cleanliness and overall condition of class influences the overall quality of education a student receives during their attendance at the school. The findings from this research indicate that to improve student academic performance and engagement while attending school, teachers should promote cleanliness and organizational skills as part of the classroom management process.

In order to successfully manage a classroom, teachers need to know what influences students' behaviour and how to respond correctly to that influence. Students' behaviour is affected by psychological, social and environmental factors that teachers need to identify to be able to react correctly (Dagdag and Bandera, 2021). Furthermore, the results of this study indicate that it is not enough to just respond to inappropriate behaviour but that teachers also require an understanding of behaviours' root causes in order to promote student engagement and lessen classroom disruptions. A comprehensive meta-analysis involving over 93,000 participants found that student engagement is strongly affected by both internal factors, such as motivation,

and external factors, including teacher behavior and classroom environment quality (Li & Xue, 2023). Also, positive teacher-student interactions and organized, supportive environments were shown to significantly enhance engagement, while negative behaviors and poor environmental conditions were linked to reduced participation and learning outcomes. These studies collectively emphasize that effective management must integrate behavioral understanding with conducive learning environments, highlighting the need for interventions that support teachers in maintaining order and maximizing student engagement.

Recent studies have developed automated classroom behavior recognition systems that can assist teachers in monitoring student behavior efficiently, reducing reliance on manual observation and workload of teachers. Integrating artificial intelligence in classrooms has the potential to transform traditional learning and management practices by automation (Li et al., 2023). Computer vision can identify student actions, allowing teachers to monitor learning environments more efficiently while being able to instruct lessons. However, the adoption of these technologies also presents challenges and areas for further research, such as ensuring reliability, ethical use, and seamless integration into everyday classroom workflows (Dimitriadou & Lanitis, 2023).

A study developed by Lin et al. (2021) uses computer vision techniques to detect student actions in real time. The system achieved 15.15% high precision and 12.15% high recall, indicating accurate reading of behaviors occurred. Another study by Yang (2024) introduced a dual-stream classroom behavior capture and analysis system that integrates AlphaPose keypoint detection with image data to monitor student behavior in online teaching environments, achieving a 97.3% accuracy on student behaviors indicating its potential for behavior analysis. These advancements suggest such technologies can support teachers in monitoring classroom

dynamics more effectively, allowing them to focus attention on instruction and overall classroom management rather than manual behavior observation.

Computer vision involves the challenge of detecting cluttered objects in an environment where the objects can overlap one another, have no fixed form, or have limited context, all of which lead to a significant decrease in the accuracy of the visual recognition system's detection ability. According to Bashkirova et al. (2021), the lack of appropriate datasets or models that can accommodate heavily cluttered scenes is a limitation to existing methodologies because of the researchers' use of a dataset that was designed to make researchers aware of the limitations of their methods to detect 'real-world' objects in real-world conditions. The same challenges that exist in detecting cluttered objects in the environment are also applicable to the classroom setting, where clutter from scattered items (ex. paper, trash, learning materials) can result in a negative impact on classroom cleanliness and organization. Liu et al. (2024) developed a model to enhance the performance of object detection by using advanced object detection techniques, such as attention mechanisms and multi-scale feature fusion, to achieve an mAP increase of 4.7% while maintaining real-time performance at an average of 39 FPS (frame/s). Although there has been much improvement, there are still deficiencies in the ability to detect randomly distributed overlapping items and visually similar entities; thus, additional research is needed to create more effective systems for the classroom environment to monitor cleanliness and manage classrooms effectively.

Recent advancements in deep learning and computer vision have enabled real-time monitoring of classroom behavior, providing teachers with continuous and objective data on student activities. Real-time detection and classification of various student behaviors have been demonstrated with a vision-based deep neural network system that has demonstrated that

automated monitoring will eliminate the requirement for constant manual observation (L. Lin et al., 2024). An improved YOLOv8-based computer vision model specifically designed for classroom environments demonstrated the ability to achieve 95% accuracy in the real-time detection of student behavior, confirming that the implementation of automated monitoring within an educational environment is a feasible option (Sheng et al., 2025). These studies provide a solid foundation for integrating automated behavior detection with a real-time alert system, making it possible for teachers to receive immediate notifications of disruptive behaviors or critical incidents. Together, such an integration supports a classroom management system that does more than simply monitor student behavior; it also provides active information to teachers that can improve engagement, discipline, and education.

Automated systems for analyzing student behavior in classroom settings have gained increasing attention as tools to support traditional observation methods and enhance instructional effectiveness. A vision-based behaviour tracking system has shown that the use of pose estimation and deep learning is capable of automatically detecting engaged or disengaged behaviours by the use of AI models to be capable of objectively interpreting ,in real-time, behaviours within the classroom environment (Alruwais & Zakariah, 2023) Traditional classroom management research has also reaffirmed the value of structured procedures, consistent expectation and reinforcement methodologies, in promoting engagement and reducing disruption (Kayhan, 2022). Similarly, there has been considerable promise in using smart classroom management methods that leverage IoT sensors and machine learning to assist teachers by monitoring conditions of their environment and students in a real-time management framework (Djami, 2022). Collectively, these referenced studies suggest that effective behaviour monitoring and managing practices have the potential to enhance engagement and improve

student outcomes; however, there is an ongoing need for an automated system that integrates the automated detection of behaviours with the educational strategies to effectively and collegially support teachers in their practise.

Although research has examined various aspects of classroom management, significant gaps remain in understanding how teachers navigate the complex demands of behavior management, student engagement, and instructional delivery in real classroom contexts (Marquez & Oropa, 2025). Research has also identified challenges that teachers face when it comes to managing students' disruptive behaviours, keeping students engaged in learning, and communicating with students in both online and face-to-face environments. However, the literature contains few examples of how technology and pedagogy can be combined to provide holistic solutions for these problems (Adsız & Dinçer, 2025). While there has been some investigation of AI-supported systems that capture and analyze student behavior, little empirical evidence exists regarding how these systems can be utilized in conjunction with teachers' regular teaching practices to support their ongoing decision making and decrease their cognitive load (Ling, 2022). Further, there are limited studies that combine elements of classroom management with detecting displaced objects. Therefore, there is a need for studies to examine classroom-specific challenges related to managing student behaviors, and to develop and evaluate combination systems for monitoring student behavior, supporting teachers' instruction, and providing usability for teachers in order to improve the overall effectiveness of the classroom and educational outcomes for students.

Disruptive behavior remains to be an ongoing issue for teachers, as they must provide instructional delivery with continuous behavior monitoring simultaneously. Teachers manually observe student behavior while teaching, which can delay or overlook necessary responses to

students when demonstrating disruptive behavior, as well as maintaining the cleanliness of the classroom. With the advancement of computer vision and artificial intelligence, there is a need for an ethical and practical system that can automatically detect students doing disruptive behaviors, capture event-based visual evidence, monitor any scattered objects that could affect learning, and provide timely support to teachers while ensuring student privacy.

This study seeks to answer the following research questions:

1. How can a computer vision-based behavior detection system be designed and developed to identify disruptive student behaviors in classrooms?
2. How accurate is the system in detecting disruptive behaviors, and scattered objects in real time?
3. How effective is the system in capturing event-based images that support teachers in reviewing detected behaviors?
4. How timely are the alerts generated by the system in assisting teacher intervention during classroom activities?
5. How can privacy and ethical considerations be ensured in the capture and storage of images related to disruptive events without identifying individual students?

The purpose of this study is to design and develop a computer vision-based behavior detection system capable of identifying disruptive and noisy student actions within a classroom setting. To achieve the main goal, the following objectives have been established:

- To develop a real-time detection model: Create a system using computer vision to accurately identify disruptive behaviors as they happen.
- To integrate an automated cleanliness monitor: Incorporate a feature that detects scattered objects or clutter to help maintain an organized learning environment.

- To build an event-based evidence capture: Build a function that captures visual evidence of specific disruptive events to assist teachers in reviewing student behavior.
- To implement a timely alert system: Ensure the system provides rapid notifications to support teachers in making immediate interventions during class activities.
- To establish ethical data protocols: Develop a method for capturing and storing images that prioritizes student privacy and follows ethical guidelines without identifying individuals.

This study addresses the challenges of classroom management by integrating technology to support both teachers and students. Using the Triple Bottom Line framework and aligning with Sustainable Development Goals (SDGs), the importance of the study is presented in terms of social, economic, and environmental impact.

The main benefit of this system is the improvement of the classroom environment for both teachers and students. It aids teachers in classroom management by reducing manual observation, especially in large classes or in situations where students have varying attention spans. By responding to disruptive behaviors, the system aims to create a more focused and productive learning environment, improving students' learning experience. In addition, the system is designed to address ethical concerns by collecting event-based data without identifying students and thus protecting their data privacy.

The proposed system improves efficiency and productivity within educational environments. Automated monitoring allows teachers to allocate more time to providing lessons rather than behavior management and administrative tasks, increasing productivity and reducing workload. Because effective classroom management correlates with enhanced academic achievement, the system could potentially foster improved student outcomes over an extended

period. Furthermore, the implementation of AI and computer vision presents a financially advantageous approach, diminishing the requirement for staff dedicated to classroom oversight.

Although primarily software-based, the system aids in preserving a tidy and structured learning environment. Through the detection of misplaced objects and classroom clutter, it bolsters school efforts focused on cleanliness and organization. Consequently, this promotes student discipline and accountability in upholding their environment, thereby cultivating a more favorable and agreeable learning environment.

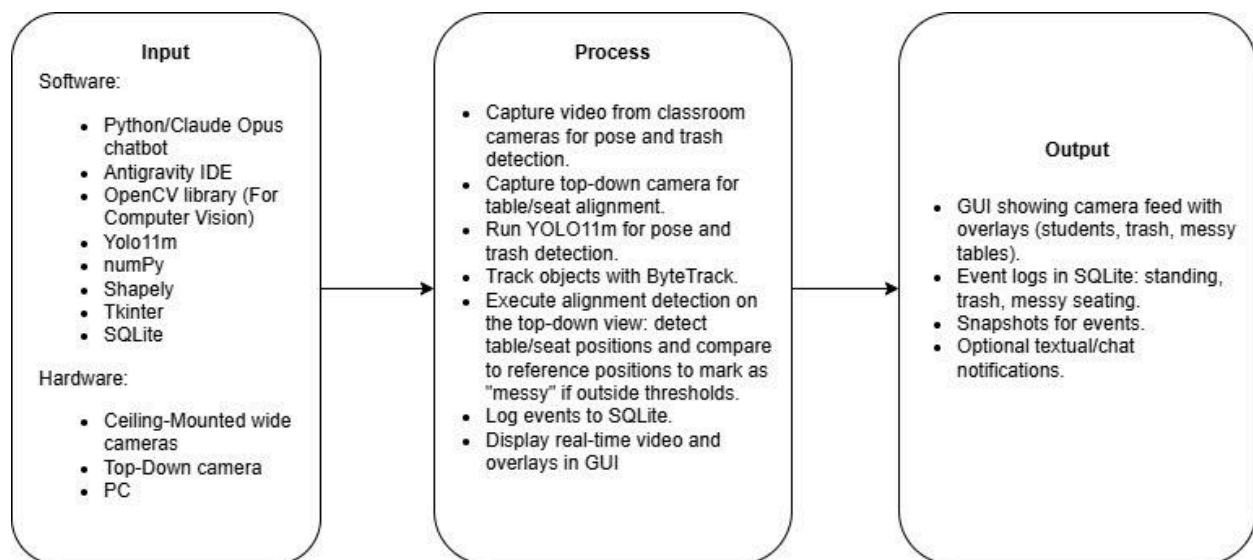
This study aligns with the Sustainable Development Goals (SDG), particularly SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure). It supports SDG 4 by improving classroom management, which is vital for improving student engagement and academic performance. Moreover, by introducing artificial intelligence and computer vision technologies into education, the study contributes to SDG 9 by fostering innovation and modernizing educational infrastructure.

The scope of this research covers the design and technical development of the BantayAralan system, which includes features for real-time behavior detection, and clutter identification. The system is developed to support teachers in managing classroom dynamics by providing automated alerts and capturing event-based images as visual evidence for behavioral review. This study is specifically intended for implementation within public grade school classrooms, focusing on the interactions between the technology and the educators. By automating these observations, the project aims to reduce the manual workload of teachers during instructional delivery. The study is scheduled for implementation and evaluation during the early months of 2026.

However, the study is subject to several limitations and exclusions to maintain a focused research approach. The system is not designed to replace the teacher's authority or decision-making but serves strictly as a support tool for classroom management. It excludes long-term psychological profiling of students, focusing instead on immediate, event-based disruptive actions. Technical constraints include the potential for reduced detection accuracy in highly cluttered environments or when objects overlap significantly. Finally, the study is constrained by strict ethical requirements, ensuring that captured data is used for review without identifying individual students to protect their privacy.

Figure 1

Conceptual Framework

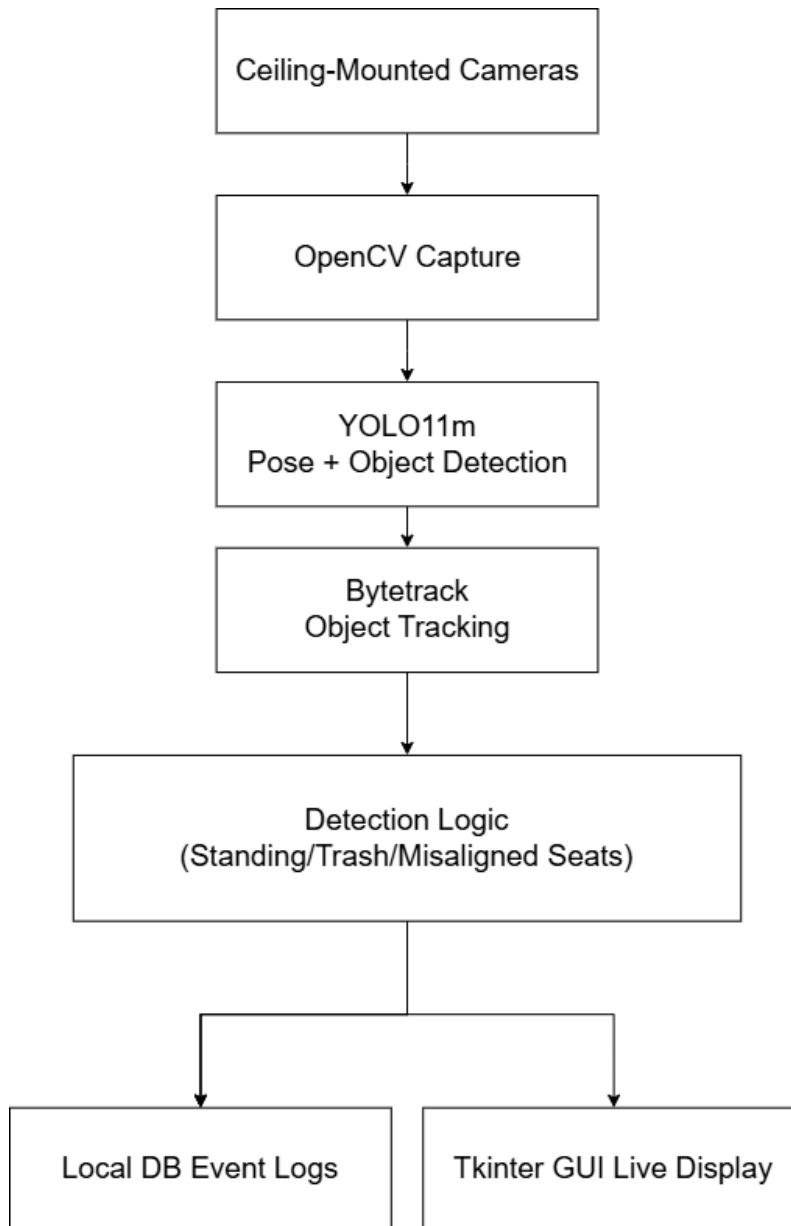


The conceptual framework presents the system using an Input–Process–Output (IPO) model. Inputs consist of hardware components such as ceiling-mounted and top-down cameras, and software tools including Python, OpenCV, YOLO11m, and supporting libraries. The process involves capturing video streams, performing pose estimation and object detection, tracking objects, and analyzing seat and table alignment using defined thresholds. These processes

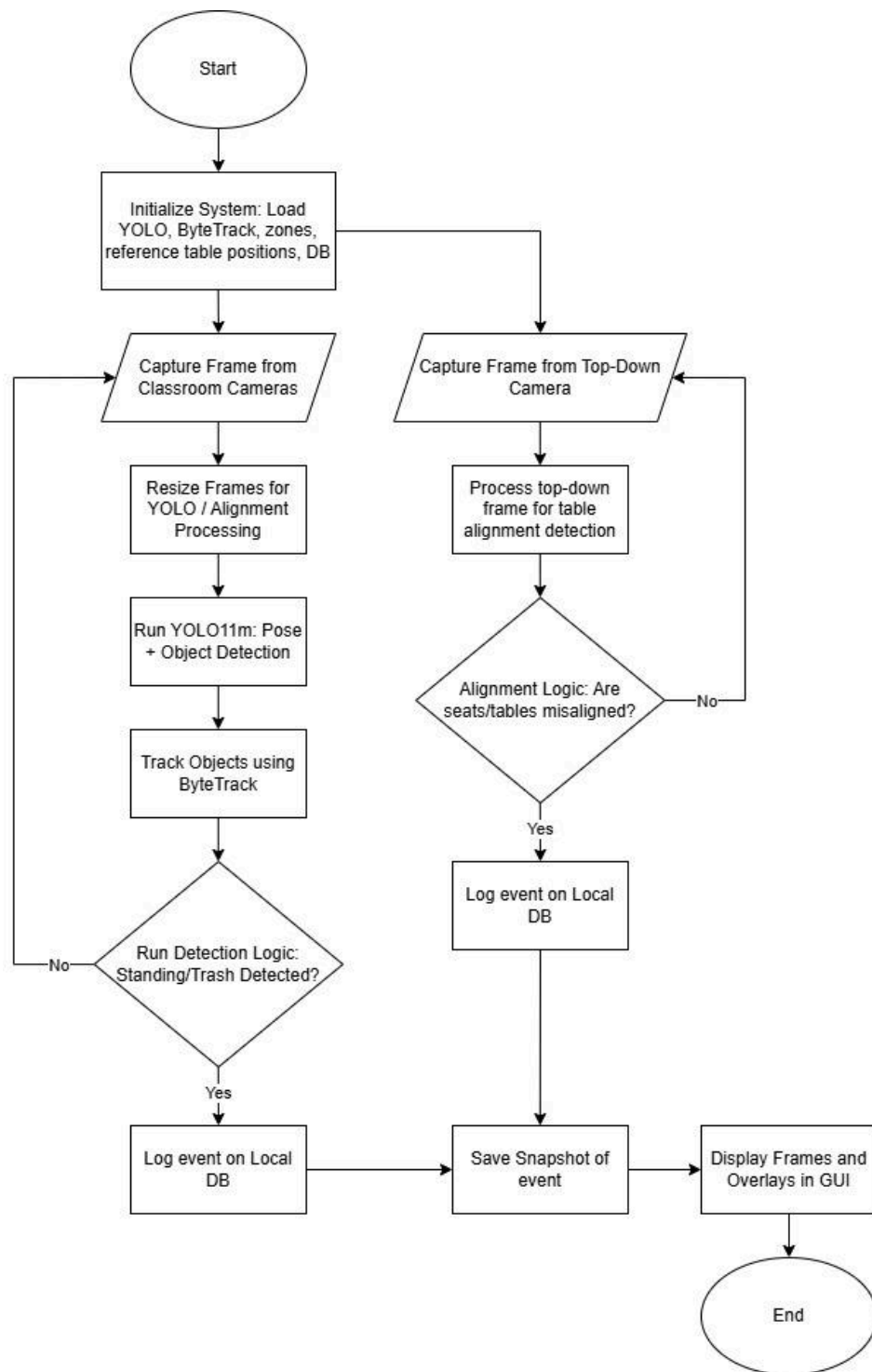
generate outputs such as real-time annotated video in the GUI, logged events (standing behavior, trash detection, and misaligned furniture), and saved snapshots, demonstrating how raw data is transformed into meaningful monitoring information.

Figure 2

Block Diagram



The block diagram shows the overall structure and data flow between system components. Video input from ceiling-mounted cameras is processed through OpenCV, YOLO11m, and tracking modules to detect behaviors and objects, while the top-down camera independently feeds into the alignment detection module for seat and table arrangement analysis. The outputs from both pipelines converge into the event logging system using SQLite and are simultaneously displayed in the desktop GUI. This diagram emphasizes the modular design and how data moves between hardware, processing units, and output interfaces in real time.

Figure 3***System Flowchart***

The system flowchart illustrates the step-by-step execution of the application, starting with system initialization and continuous frame capture with both classroom and top-down cameras. Each frame is analyzed using YOLO11m to find objects and ByteTrack to follow them, then decisions are made about behaviors (like whether someone is standing or if there is trash) and alignment. Based on these decisions, events are logged, optional snapshots are saved, and processed frames are displayed in the GUI. The flow repeats in a loop for real-time monitoring, clearly showing the control flow, processing stages, and decision points within the system. The system flowchart illustrates the step-by-step execution of the application, starting from system initialization and continuous frame capture from both classroom and top-down cameras. Each frame undergoes processing through YOLO11m for detection and ByteTrack for tracking, followed by decision-making stages where behaviors (standing, trash presence) and alignment conditions are evaluated. Based on these decisions, events are logged, optional snapshots are saved, and processed frames are displayed in the GUI. The flow repeats in a loop for real-time monitoring, clearly showing the control flow, processing stages, and decision points within the system.

Methods

Research Design

This study adopts a Developmental Research Design (also known as Design and Development Research or DDR) integrated with a Quantitative Experimental Evaluation. This combined approach is appropriate because the study does not merely observe phenomena but creates, implements, and evaluates a technological solution, the BantayAralan system (Richey & Klein, 2005; Wang, 2025). Developmental research is specifically suited for studies that aim to design, build, and assess innovative systems or models, particularly in engineering and

technology-related fields, as demonstrated in studies developing instructional design practices (Richey & Klein, 2005), IoT-enabled smart classroom management frameworks (Wang, 2025), and real-time classroom air quality monitoring systems (Rawat et al., 2025). In this study, the primary goal is to develop a monitoring system for classroom behavior based on computer vision, making the developmental approach the most appropriate methodological choice, akin to DDR applications in recent educational technology prototypes (Wang, 2025). To complement this study, a quantitative evaluation component is incorporated to objectively measure the system's performance using numerical indicators such as accuracy, detection rate, response time, and system efficiency, mirroring metrics like usability scores, system effectiveness, and environmental parameter reductions in prior tech development research (Rawat et al., 2025; Wang, 2025). This ensures that the evaluation is objective, replicable, and statistically analyzable (Richey & Klein, 2005).

Procedure

The study will be conducted through four major phases, system development, system testing, system implementation, and evaluation, which aligns according to the standard Developmental Research Design (DDR) procedures that emphasize iterative design, prototyping, testing, and validation (Sahrir et al., 2012; Jaya et al., 2021). In the system development phase, researchers will design and develop the BantayAralan system, integrating hardware like cameras with Python-based computer vision models for detecting disruptive behaviors, clutter, event-based imaging, real-time alerts, database logging, and a graphical user interface (GUI), similar to hardware-software integration in IoT smart classroom prototypes (Jaya et al., 2021; Smart Classroom Control System Design, 2024). This is followed by a testing phase using controlled scenarios of normal, disruptive, and unclean classroom conditions to measure

detection accuracy. The response time, stability, and reliability were assessed through multiple runs with refinements, mirroring unit testing and scenario validation in DDR for educational technologies (Sahrir et al., 2012; Smart Educational Environment Monitoring, n.d.). After testing, implementation occurs in actual or simulated classrooms with optimized camera positioning for real-time monitoring, alerts, imaging, and teacher orientation, consistent with real-setting deployment in DDR frameworks (Jaya et al., 2021; Smart Classroom Control System Design, 2024). Finally, evaluation involves pre-implementation surveys on teachers' classroom management challenges and needs, followed by post-implementation surveys on usability and effectiveness, supplemented by system-generated data like logs, timestamps, images, and metrics (accuracy, latency, speed), using formative and summative methods akin to those achieving high accuracies in comparable studies (Sahrir et al., 2012; Smart Educational Environment Monitoring, n.d.).

Instruments/Tools

The study will make use of hardware and software, as well as data collection instruments in developing and evaluating the BantayAralan system. These hardware and software tools play an important role in data collection and system evaluation.

The study will utilize essential hardware components such as cameras and a computer system to collect data necessary for monitoring classroom conditions in real time. The following are the components to be used:

- Ceiling-mounted camera – captures student behavior across the classroom
- Top-down camera – detects clutter and monitors desk alignment
- Computer/Laptop – processes video input and runs the system

The study will make use of a combination of programming tools and frameworks to develop, process, and analyze data, which will allow for accurate detection, tracking, and evaluation of activities in the classroom through the BantayAralan system. The following are the components to be used:

- Python – main programming language
- OpenCV – video and image processing
- YOLO-based model (e.g., YOLO11m) – object and behavior detection
- ByteTrack – object tracking
- SQLite – database for event logging
- GUI framework (e.g., Tkinter or PyQt) – user interface

The primary method for data collection will be through the administration of a structured survey questionnaire to the participating teachers. The questionnaire will have two sections. One will be administered prior to the actual use of the system, while the other will be administered after the actual use. The first part will aim to measure the need for a behavior management system by examining teachers about their difficulties in monitoring students, dealing with behavioral problems, keeping the classroom clean, and the difficulties associated with manual observation. The second part will aim to measure the usability and effectiveness of the BantayAralan system, including ease of use, system clarity, detection accuracy, promptness, and overall usefulness. The scale will range from Strongly Disagree (1) to Strongly Agree (5) using a 5-point Likert scale with five points.

Apart from the survey, the system will also function as a data collection instrument through the production of quantified data. This includes detected behavior, detected objects, time stamps of raised alerts, logs of events, images captured, and system performance data such as

accuracy, precision, recall, frames per second (FPS), and latency. This data will represent objective measures of system performance.

To ensure the validity of the instrument, the questionnaire will be reviewed by selected experts, such as teachers or IT professionals, to establish its clarity, appropriateness, and relevance. This will ensure that it is adequately modified prior to its administration to ensure that it adequately captures the necessary information for the research.

Sources of Data

The primary source of data used in this research is the qualitative interviews and surveys carried out on teachers. The qualitative interviews aim to obtain data on the experiences of teachers in handling classroom management issues like controlling disruptive behaviors among students and ensuring classroom cleanliness, while the survey aims to obtain data on how effective and useful the BantayAralan system is. The data used in this research will be primarily on the experiences of teachers in handling student engagement and how useful the BantayAralan system is in a real-world scenario. The research will also include system-generated logs and metrics on how accurate and fast the computer vision model is in its responses as a secondary data source. The secondary data used in this research includes gathering related literature and studies on a particular topic that is related to this research, such as classroom management and computer vision.

Participants

The participants in this study will include public grade school teachers, as they are the ones that manage the classroom and the students. These individuals are selected because they are the ones that are challenged by the task of managing the classroom and the students. They will determine the system's ability to effectively observe the student's behavior and the role that the

system plays in the decision-making process. The researchers will utilize a purposive sampling method because the participants are involved in the management of the classroom and student instruction. All participants will be informed to ensure the system design continues to be helpful and respects the ethical standards of the classroom environment.

Data Analysis Plan

The data collected from teacher interviews and survey questionnaires will be grouped based on recurring themes related to system functionality, usability, reliability, and performance. The responses will be used to collect insights on classroom management and determine the effectiveness of the BantayAralan system. The survey questionnaire will also be used for teacher feedback. The questionnaire will use a 5-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5). Each of the responses will be reviewed to identify patterns in teacher experience and system effectiveness. Frequency of similar responses will be recorded to support consistency in findings. Additionally, data generated from the system will also be collected which shows how the system detects behavior, tracks objects, and responds to events. The results will be interpreted to determine overall system acceptability and identify areas for improvement, as well as determining if objectives of the study are met and accomplished.

Standard Set

The researchers used internationally recognized standards to measure software quality and system performance. Such standards offer specific parameters to measure system accuracy in terms of functionality, speed in terms of processing, usability in terms of interfaces, and artificial intelligence in terms of computer vision. The selected standards include ISO/IEC 25010 for software quality characteristics, ISO/IEC 23053:2022 for artificial intelligence system

frameworks, and ISO/IEC 25023:2016 for measurable quality indicators. Each standard provides a systematic approach to measure the system and obtain reliable results.

Table 1

Product Quality of BantayAralan (ISO/IEC 25010)

Characteristics	Sub-characteristics	Description
Functional Suitability	Accuracy	Measures how accurately the system performs required functions, ensuring correct detection and reporting of classroom behavior.
Performance Efficiency	Response Time	Assesses system speed and resource usage during video processing and AI detection, providing measurable performance metrics.
Usability	Learnability	Evaluates how easy it is for teachers to navigate the interface and interact with system features without extensive training.
Reliability	Fault Tolerance	Measures the system's ability to operate under normal and unexpected conditions, maintaining continuous behavior monitoring.
Security	Data Protection	Verifies the system's ability to protect student data against unauthorized access and data breaches.

ISO/IEC 25010 is a standard that describes a product quality model that divides software into various quality characteristics and sub-characteristics. These are the qualities that determine the level of satisfaction that the software system provides in meeting the requirements and expectations of the user. The qualities include functional suitability, performance efficiency,

usability, reliability, security, compatibility, maintainability, and portability. These are the specific criteria that the software system is evaluated against. This standard provides a basis for evaluating the functions of the software in terms of accuracy, the response of the system, the user interface, and the stability of the system. In the case of the BantayAralan System, the ISO/IEC 25010 standard provides a basis for evaluating the functions of the system in terms of user interaction and the quality of the system (*ISO 25010*, n.d.).

Table 2

Quality Measurement Focus Areas of BantayAralan (ISO/IEC 23053:2022)

Characteristics	Sub-characteristics	Description
Functional Suitability	Detection Accuracy	Quantifies the percentage of classroom behaviors correctly identified by the AI component.
Performance Efficiency	Processing Speed	Measures average frame processing time, CPU load, and memory usage during system operation.
Reliability	Failure Rate	Counts system failures during extended operation to evaluate consistency and stability.
Resource Utilization	System Load	Assesses computational resource use during AI detection tasks, ensuring efficient operation.

ISO/IEC 23053:2022 establishes a framework and terminology to express artificial intelligence systems using machine learning technology. The standard establishes the fundamental components, functional requirements, and processes of machine learning systems. The standard allows evaluators and developers to have a shared understanding of the behavior of

the system, the training of the models, and the system's integration. The standard is applicable to software that receives input from cameras, recognizes behavior patterns, and classifies the behavior. For the Classroom Behavior Management System, ISO/IEC 23053 provides guidelines on how to express the documentation of the AI system, the training data, and the outputs of the system, ensuring that all aspects of the system's evaluation are consistent in their measurement and verification of the detection and the integration of the system into the software interface (ISO/IEC 23053:2022, 2025).

Table 3
Prototype Quality of BantayAralan (ISO/IEC 25023:2016)

Characteristics	Sub-characteristics	Description
AI Model	Training	Details the dataset preparation, training process, and model updates for behavior recognition.
AI Model	Output Accuracy	Evaluates how well the model predicts or classifies classroom behaviors based on video input.
System Integration	Interface Compatibility	Assesses how the AI module communicates with the main software, including input handling and output display.
Lifecycle Management	Version Control	Tracks updates to AI models, software patches, and system improvements to maintain performance consistency.

ISO/IEC 25023:2016 offers quantitative metrics for assessing software quality attributes, as delineated in ISO/IEC 25010. This standard furnishes metrics applicable to both internal and external system characteristics, thereby facilitating an objective evaluation of software

performance. It encompasses measurements including response time, failure rate, resource utilization, and processing efficiency. Regarding the Classroom Behavior Management System, this standard furnishes quantifiable indicators for AI detection performance, system processing speed, and overall reliability. Metrics such as average detection time, accuracy, and CPU or memory consumption provide tangible evidence of the system's efficacy. Consequently, the application of ISO/IEC 25023 enables evaluators to quantify software quality and compare outcomes against established benchmarks, thereby yielding dependable and replicable assessments (*ISO/IEC 25023:2016*, n.d.).

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